PASINDU INDUWARA

Software Engineer

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PROFILE

Software Engineer and active open source contributor with a passion for web development and building scalable solutions. Industry experience as a Software Engineer Intern at WSO2, where I specialized in building Internal Developer Portals (IDP) using Backstage and OpenChoreo. Experienced in designing full-stack applications, cloud deployments, and modern DevOps practices. Skilled in translating user needs into functional interfaces while adhering to clean code standards. Proficient in multiple technology stacks and committed to continuous learning and open source collaboration.

WORK EXPERIENCE

SOFTWARE ENGINEER INTERN | WSO2 | JULY 2025 - DECEMBER 2025

Engineered a full-stack plugin for Backstage that integrated WSO2 Choreo's deployment workflows, significantly streamlining application provisioning via the Software Catalog. As a core contributor to the open source Internal Developer Platform OpenChoreo, developed a scalable Incremental Entity Ingestion module to optimize Kubernetes resource synchronization for enterprise-scale systems. Furthermore, eliminated critical API bottlenecks by implementing cursor-based pagination, which successfully resolved timeout issues during high-volume data processing and ingestion cycles.

OPEN SOURCE CONTRIBUTIONS

- **SST (Serverless Stack)** - Active contributor to the models.dev repository. Integrated multiple AI providers and configured high-demand models, optimizing both backend configurations and frontend UI consistency.
- **KiloCode** - Contributed to documentation improvements regarding model-initiated slash commands and editor interactions.

EDUCATION

UNIVERSITY OF KELANIYA | 2023 - PRESENT | EXPECTED GRADUATION - OCTOBER 2026

- B.Sc. Honours in Information Technology, Current GPA: 3.7

NIWATHTHAKACHETHIYA NATIONAL COLLEGE | 2012 - 2019

PROFESSIONAL QUALIFICATIONS

TRAINEE FULL STACK DEVELOPER | University of Moratuwa | Centre for Open & Distance Learning

Completed full-stack developer training covering Python programming, web development, and professional practice. Gained hands-on experience in frontend and backend technologies through Python Programming, Web Design, and Server-Side Development courses. Emphasized real-world problem-solving, modern software tools, and soft skills including communication, teamwork, and project management.

PERSONAL PROJECTS

1. Guidia – Full-Stack Career Guidance Platform (Individual Project)

Developed a comprehensive career guidance platform to digitize the University of Kelaniya's recruitment and counseling processes. Built a robust full-stack solution using React, TypeScript, and Zod for the frontend, backed by an Express and MySQL architecture with Socket.IO for real-time messaging. Deployed the application on Azure Container Apps with Azure Blob Storage, integrating multi-AI services to connect students, counselors, and companies efficiently.

2. Browtrix – MCP Browser Automation Ecosystem (Individual Project)

Architected a real-time AI automation ecosystem using the Model Context Protocol (MCP) to bridge AI assistants with web browsers. Engineered a high-performance backend using Python FastMCP, FastAPI, and WebSockets within a Turbo monorepo to enable complex human-in-the-loop workflows. Built a cutting-edge frontend with Next.js 15, React 19, and Tailwind CSS 4, featuring HTML snapshotting, interactive confirmation modals, and connection pooling for seamless automation.

3. Quota.app - Fuel Quota Management System (Group Project)

Designed and deployed a microservices-based fuel management system connecting vehicle owners, stations, and administrators. Developed a Spring Boot backend with JWT authentication and JPA, integrated with three Next.js frontends and an Expo/React Native mobile app. Implemented QR-based transaction processing, SMS notifications via Twilio, and PostgreSQL reporting to handle real-time quota tracking and fuel distribution.

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4. Personalized Ad-Copy Generation – AI/NLP Project (Group Project)
Engineered a RAG-based ad generator using T5, FAISS, and LangChain deployed on Azure AI. Optimized NLP pipelines with FP16, gradient accumulation, and dynamic GPU memory for high-performance inference.

5. Multi-Link Sharing Platform ("Linky") (Group Project)
Cloud-native platform for personalized landing pages with multiple external links. Deployed on Azure using Kubernetes and Docker with MongoDB Atlas. Ensured high availability, load balancing, and TLS encryption.

6. Hela Rasa Recipe Android Mobile Application (Group Project)
Android app for managing and sharing recipes with user login, multimedia-supported recipe creation/editing, and social sharing. Designed UI/UX with wireframes, used Firebase for cloud storage with thorough testing.

7. Automated QA Testing Framework (Group Project)
Developed automated testing framework using Selenium, TestNG, and Eclipse for cross-browser testing. Implemented testing techniques including Equivalence Partitioning, Boundary Value Analysis, and Decision Table Testing. Executed automated test suites via Azure DevOps CI/CD pipelines with Postman for API testing and generated comprehensive test reports.

ACHIEVEMENTS

First Runner-Up

- J'PURA EXPO 2023, Inter University Export-Oriented Innovation Competition.

Semi-Finalist

- Achieved Semi-finalist status for the project "Ceylon Treasure" in the VentureVerse Startup Challenge conducted by Sabaragamuwa University

VOLUNTEER WORK AND AFFILIATIONS

- **Assistant Media Director**, Industrial Management Science Students' Association (IMSSA), 2024 - 2025
- **Member**, AIESEC Colombo North Local Committee, 2023 - 2025
- **Member**, Gavel Club - University of Kelaniya, 2023 - 2025
- **Volunteer**, Sasnaka Sansada Foundation, 2022 - 2024

SKILLS

Tech Skills

- **Programming Languages:** HTML, CSS, JavaScript, TypeScript, Python, Java, C/C++, SQL, XML
- **Frameworks & Libraries:** React, Next.js, Tailwind CSS, Node.js, Express.js, TS-Rest, Angular, Spring Boot, Flutter, React Native, Expo, Java Servlets, JSP, jQuery, shadcn/ui, React Hook Form, Zod, Recharts, Socket.IO, FastMCP, FastAPI, WebSockets
- **Tools & Platforms:** MySQL, PostgreSQL, MongoDB, Docker, Kubernetes, Azure (Container Apps, Blob Storage, AI Services, DevOps), Firebase, Git, Maven, Apache Tomcat, Vite, Sentry, Axios, Figma, Supabase, Twilio, Faker, Hoppscotch, Eclipse, Backstage.io, WSO2 Choreo, OpenChoreo
- **Testing & QA**: Selenium, TestNG, Postman, API Testing, Automation Testing, CI/CD Test Integration, Cross-browser Testing
- **AI/ML Technologies**: LangChain, FAISS, Transformers, PyTorch, spaCy, Pandas, Sentence Transformers, Azure AI Services
- **Web Development**: Responsive Design, UX/UI, Cross-Browser Compatibility, RESTful APIs, WebSocket Services, Authentication (JWT, OAuth, OTP), Real-Time Messaging, Cloud Deployment (Azure)

Soft Skills

- Adaptability
- Creativity
- Critical Thinking
- Effective Communication
- Emotional Intelligence
- Lifelong Learning
- Problem Solving
- Team Collaboration
- Time Management

LANGUAGES

- English (Fluent)
- Sinhala (Native Speaker)

REFERENCES

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